Full-Waveform Inversion for Ring-Array Ultrasound: Algorithm, Simulation Setup, and Abdominal CT Results

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Abstract

This note describes the time-domain full-waveform inversion (FWI) implemented in `invert_ring.py`, which recovers a two-dimensional sound-speed map from ring-array ultrasound traces, and `totalseg_to_speed.py`, which builds abdominal sound-speed maps from an open, expert-segmented CT dataset and a table of tissue acoustic properties. On a TotalSegmentator abdomen (1450–3750 m/s, 128 elements), a two-stage inversion (10–50 kHz on a 4.0 mm grid, then 50–125 kHz on a 1.83 mm grid, with total-variation regularization) started from uniform water recovers the organs, fat layers and muscle in 6.4 min on an RTX A6000. More than 10 mm from bone the soft-tissue RMS error is 5.7 m/s (liver 7, kidney 6, muscle 8, fat 8 m/s); the remaining error sits in a thin layer around bone, and bone itself is located but too slow (median 2730 against 3750 m/s). A third stage at 100–250 kHz, run from this model, cycle-skips and destroys the reconstruction regardless of earlier settings, so the two-stage model is the result and the fine stage is left to future work.

1 Forward model

The forward model is the 2D constant-density acoustic wave equation, discretized with second-order finite differences in space and time (FDTD). Sound speed $c(\mathbf{x})$ is the only material property; density and attenuation are constant:

$$\frac{\partial^2 u}{\partial t^2} = c(\mathbf{x})^2 \, \nabla^2 u + s(t) \, \delta(\mathbf{x} - \mathbf{x}_{\mathrm{tx}}). \tag{1}$$

- **Time stepping.** A leapfrog scheme advances the pressure, $u^{n+1} = 2u^n - u^{n-1} + \Delta t^2 c^2 L[u^n]$, where L is the 5-point Laplacian applied as a 3×3 convolution.
- **Stability.** The step obeys the 2D CFL bound $c_{\max} \, \Delta t \, \sqrt{1/\Delta x^2 + 1/\Delta y^2} < 1$. The code uses a CFL number of 0.45, with $c_{\max}$ the larger of the true medium maximum and the upper reconstruction bound.
- **Boundaries.** First-order Mur absorbing conditions on all four edges; corners average the x- and y-direction updates.
- **Sources and receivers.** A transmitter is a soft (additive) point source at its nearest grid node. Receivers sample pressure by bilinear interpolation, so element positions are not locked to pixel centres.
- **Execution.** Fields stay on one PyTorch device (CPU, CUDA or Apple MPS). Independent transmitter shots are advanced together as a batch, and on CUDA the step is compiled with `torch.compile`.

2 Simulation setup

A ring of point transducers surrounds the object; each element both transmits and receives. One element fires a chirp per shot and every element records, giving an $N \times N \times$ time dataset. Two configurations are used: the default phantom setup (200 mm field of view, 70 mm ring) and the abdominal setup, which scales the grid and ring to fit a 0.3 m-wide torso.

2.1 Acquisition and discretization

Table 1: Simulation parameters. Abdomen values (TotalSegmentator subject s1366) are listed per multiscale stage; the optional third stage is in Section 7.

Parameter	Phantom setup (default)	Abdomen setup, per stage
Grid (cells)	401×401	$101^2 \rightarrow 201^2$
Grid spacing Δx	0.50 mm	4.00 $\rightarrow$ 1.83 mm
Field of view	200 mm	404 $\rightarrow$ 367 mm
Ring radius	70 mm	167.7 $\rightarrow$ 168.3 mm
Elements	256 (pitch 1.7 mm)	128 (pitch 8.3 mm)
Coupling medium	water, 1500 m/s	water, 1480 m/s
Source pulse	Hann-windowed linear chirp, 100–250 kHz, 20 μs	10–50 kHz / 80 μs $\rightarrow$ 50–125 kHz / 40 μs
Time step Δt (CFL 0.45)	$\approx$ 80 ns	326 $\rightarrow$ 149 ns
Record length	1.6$\times$ ring diameter / slowest speed	$\approx$ 371 μs (1135 $\rightarrow$ 2490 steps)
Points per shortest wavelength	$\approx$ 12	$\approx$ 7.3 $\rightarrow$ 6.3
Inversion bounds on c	1400–2000 m/s	1400–3900 m/s

The abdomen time steps follow from the 3900 m/s upper bound, which sets the CFL limit. The record length is 1.6 times the time to cross the ring diameter at the slowest speed in the medium (fat, 1450 m/s). The grid is never finer than the CT pixels (1.5 mm); the final stage-2 grid is 1.83 mm.

2.2 Abdominal sound-speed model from an open segmentation

The medium is one axial slice of a public, expert-segmented CT, converted by `totalseg_to_speed.py`. Organs, vessels, bowel, bone and the deep back muscles come from the segmentation; only pixels no mask covers are classified by Hounsfield units (HU).

1. **Data.** The *Small subset of TotalSegmentator Dataset* v2.0.1 [5, 4] holds 102 CT volumes resampled to 1.5 mm, each with masks of 117 structures (CC BY 4.0). `-fetch` downloads a single subject out of the 3.2 GB zip with HTTP range requests (44 MB for subject s1366).

2. **Subject choice.** Of the 54 subjects whose masks include the main upper-abdominal organs, the cleanest were screened by compressed bytes per voxel, then by the HU standard deviation inside the liver after removing a 5^3-voxel local mean. Subject s1366 has 14.7 HU of noise, against 140 HU for the noisiest (low-dose) scan, s0407, which was rejected.

3. **Slice choice.** Among slices showing all ten key structures (liver, spleen, both kidneys, pancreas, stomach, aorta, inferior vena cava, both deep back muscles), the one with the least gas inside the body is used: slice 259, at the T12/L1 level, with 2.5% gas.

4. **Tissue classes.** Masks are merged into the classes of Table 3. Unlabelled body pixels are classified by HU after a 5×5 median filter: fat (-190 to -30), gas (< -190) or other soft tissue. Gas, which the scalar fluid model cannot represent, becomes water; inside segmented solid organs low HU is treated as noise, not gas. The body is the largest connected region above -500 HU, which excludes the arms and the scanner table.

5. **Acoustic properties per class.** Each class points to one row of `abdominal_acoustic_parameters.csv` (Table 2), compiled from [1, 2, 3]. The script writes speed, density, power-law attenuation and B/A maps; the present solvers are lossless and constant-density, so only speed enters them. Gas is modelled as water rather than 343 m/s air: without a density contrast the scalar model cannot reproduce the near-total reflection at a gas boundary.

6. **Embedding.** `load_ct_ring_grid` centres the body, resamples it to each stage's grid with nearest-neighbour sampling, pads the outside with 1480 m/s water, and places the ring 10 mm outside the body with an 8-cell margin to the grid edge.

Table 2: Tissue acoustic properties (`abdominal_acoustic_parameters.csv`; values compiled from [1, 2, 3]). Attenuation follows $\alpha = \alpha_0 f^{\,y}$ with f in MHz.

Tissue	c (m/s)	ρ (kg/m^3)	α_0 (dB/MHz^y/cm)	y	B/A
Soft tissue (machine average)	1540	1000	0.5	1.1	7.0
Water / bile / urine	1480	1000	0.0022	2.0	5.0
Fat (visceral/subcutaneous)	1450	920	0.75	1.25	10.0
Liver	1570	1060	0.6	1.2	6.8
Kidney	1560	1050	0.9	1.0	7.2
Blood	1575	1050	0.15	1.3	6.1
Muscle (abdominal wall)	1580	1080	1.1	1.0	7.4
Spleen	1567	1054	0.55	1.2	6.8
Bowel gas / air	343	1.2	12.0	2.0	0.4
Bone / cortical rib	3750	1900	15.0	1.5	0.0

Table 3: Tissue classes of the s1366 slice (333×333 pixels at 1.5 mm) and the table row each uses.

Class	Source	Property row	c (m/s)	Share of body
Fat	HU	Fat	1450	31.9%
Other soft tissue	HU	Soft tissue average	1540	23.5%
Liver	mask	Liver	1570	21.8%
Digestive tract (stomach, small bowel, colon)	masks	Soft tissue average	1540	7.1%
Muscle (deep back muscles)	masks	Muscle	1580	4.3%
Gas, modelled as water	HU	Water	1480	2.5%
Bone (T11–L1 vertebrae, ribs)	masks	Bone	3750	2.3%
Kidney	masks	Kidney	1560	2.1%
Blood vessels (aorta, IVC, portal/splenic vein)	masks	Blood	1575	1.9%
Other organs (pancreas, adrenals)	masks	Soft tissue average	1540	1.8%
Spleen	mask	Spleen	1567	0.8%

The map spans 1450–3750 m/s. Soft-tissue contrasts are small (within 7% of water), while bone is 2.5 times faster than water: it dominates both the data and the difficulty of the inversion.

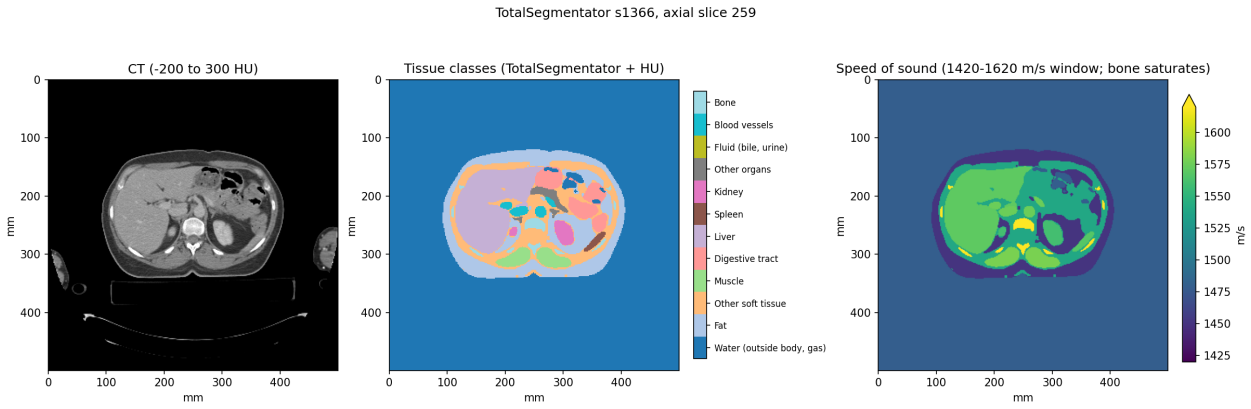


Figure 1: TotalSegmentator subject s1366, axial slice 259: CT (window −200 to 300 HU), merged tissue classes, and the resulting sound-speed map. The speed map is shown in a 1420–1620 m/s soft-tissue window; bone (3750 m/s) saturates at the top of the scale.

Earlier model. The first experiments (Section 6.3) used `ct_to_speed.py` on one slice of C4KC-KiTS case KiTS-00136 (512 × 512 at 0.684 mm). That pipeline has no anatomical labels: it thresholds HU into air, fat, soft tissue and bone, and splits superficial from deep tissue at 25 mm depth (fat into subcutaneous and visceral, soft tissue into muscle and organs), with representative speeds from Table 1 of [6] (1478–3476 m/s). The segmentation-based model replaces that depth rule with real organ, vessel and muscle boundaries.

3 Inversion algorithm

The inversion minimizes the least-squares misfit between simulated and observed traces over slowness squared $m = 1/c^2$, using an exact discrete adjoint for the gradient and first-order updates. Everything below is implemented in `invert_ring.py` and `fdtd2d/inversion/`.

3.1 Unknowns and constraints

- **Parameter.** $m = 1/c^2$ on every pixel inside an interior disk Ω of radius (ring radius − margin), with a 2-pixel margin by default. Outside Ω, c is fixed at the coupling-medium speed. The wave operator is linear in $c^2 = 1/m$, which keeps the gradient simple.
- **Bounds.** After every update m is projected onto $[1/c_{\max}^2,\ 1/c_{\min}^2]$ and the exterior is reset to the background.
- **Starting model.** Homogeneous coupling medium (1500 or 1480 m/s) inside the mask. For diagnosis only, `--start-blurred-truth` σ starts from the true slowness blurred by a Gaussian of σ mm.

3.2 Objective

$$J(m) = \frac{1}{2} \sum_{s \in S} \sum_{r \neq s} \sum_{n} \left(d_{s,r,n}^{\text{pred}}(m) - d_{s,r,n}^{\text{obs}} \right)^2 + \frac{\alpha}{2} \left\| \nabla m \right\|_{\Omega}^2 + \beta \sum_{\Omega} \left(\sqrt{|\nabla m|^2 + \varepsilon^2} - \varepsilon \right). \qquad (2)$$

S is the set of active transmitters, r runs over receivers and n over time samples. The transmitter's own channel is excluded because a point source's direct pressure is a poorly conditioned datum. (In

batched runs before commit `d519e92` this exclusion silently failed; see Section 6.1.) The Tikhonov term penalizes squared differences between neighbouring pixels inside Ω, with weight α (`--reg`). The total-variation (TV) term penalizes the magnitude of the forward-difference gradient, smoothed by ε (`--tv-eps`, default 10^{-9} in units of m, about 2 m/s at 1500 m/s), with weight β (`--tv`). Because TV grows linearly with a jump, it suppresses speckle while leaving sharp organ and bone boundaries much cheaper than the quadratic Tikhonov term does. Both weights default to 0 and can be set per stage.

Observed data are synthetic: at the start of each stage the same solver simulates all N transmitters through the true medium, without noise.

3.3 Adjoint-state gradient

For each active shot the forward pass stores the full wavefield history u^n. The adjoint field λ is then stepped backward in time with the transposed operator $L(c^2\,\cdot)$, driven by the residuals scattered onto the grid by the transpose of the bilinear receiver sampling. The gradient with respect to c^2 accumulates as

$$g_{c^2} = \Delta t^2 \sum_n \lambda^n \odot L\big[u^{n-1}\big], \qquad \frac{\partial J}{\partial m} = -\frac{g_{c^2}}{m^2} + \nabla_m R(m), \tag{3}$$

where R is the sum of the Tikhonov and TV terms. The gradient is set to zero outside the mask. With `--store-every` k the sum runs over every k-th forward step, weighted by k, and `--wave-dtype float16` stores the forward fields at half precision; on the test problem these change the gradient by 2×10^{-7} ($k = 4$) and 9×10^{-5} (float16). The adjoint reuses the forward Mur boundary update, which is not its exact transpose; the finite-difference gradient check (Section 5) bounds the effect of this.

3.4 Optimizers

Table 4: Optimizers available through `--optimizer`.

`--optimizer`	Update	Notes
`adam` (default)	Projected Adam on m, $\beta_1 = 0.9$, $\beta_2 = 0.999$	Step size `--adam-lr` in units of m ($\mathrm{s}^2/\mathrm{m}^2$), per stage: 10^{-9} default; the abdomen run uses 10^{-8} on the coarse stage and 4×10^{-9} on stage 2. Each pixel moves by at most about the step size per iteration. Tolerates the noisy gradients of shot sub-sampling.
`cg`	Polak–Ribière nonlinear CG, Armijo backtracking, restart when $\beta < 0$	Deterministic; conjugacy is lost when shots are re-sampled each evaluation.
`gradient-descent`	Projected fixed step `--step-size`	Baseline.

`--model neural` replaces the per-pixel unknowns with the weights of a Fourier-feature coordinate network; the same adjoint gradient is passed through the network by the chain rule.

3.5 Shot sub-sampling

Each gradient evaluation uses K transmitters drawn at random without replacement (`--n-shots`, default 4; `--shot-seed` fixes the sequence), so one evaluation costs $2K$ wave solves instead of $2N$.

Shots run in GPU batches of up to `--parallel-shots`. Its default, `auto`, sizes each stage's batches from free GPU memory and splits the shots into equal batches; the abdomen run fits all 32 shots in one batch.

3.6 Multiscale continuation

`--multiscale` runs coarse-to-fine stages on progressively finer grids (Table 1). Each stage regenerates its own observed data, and the final m of one stage is transferred to the next by bicubic interpolation. `--stages` sets how many run: the default of 2 runs the 101^2 and 201^2 stages, and the stage-2 model is the result. A third, 401^2 stage at 100–250 kHz is available with `--stages 3` but is experimental (Section 7). `--coarse-band` sets the first stage's chirp band (default 25–75 kHz; the abdomen uses 10–50 kHz). Starting at a low frequency lengthens the period relative to the initial travel-time errors, which reduces cycle skipping: fitting a trace one period off, which traps inversions in a wrong model. `--n-elements`, `--max-iter`, `--adam-lr`, `--reg` and `--tv` take either one value or one value per stage, and `--save-model X.npz` also saves every earlier stage as `X_stage`k`.npz`.

4 Algorithm at a glance

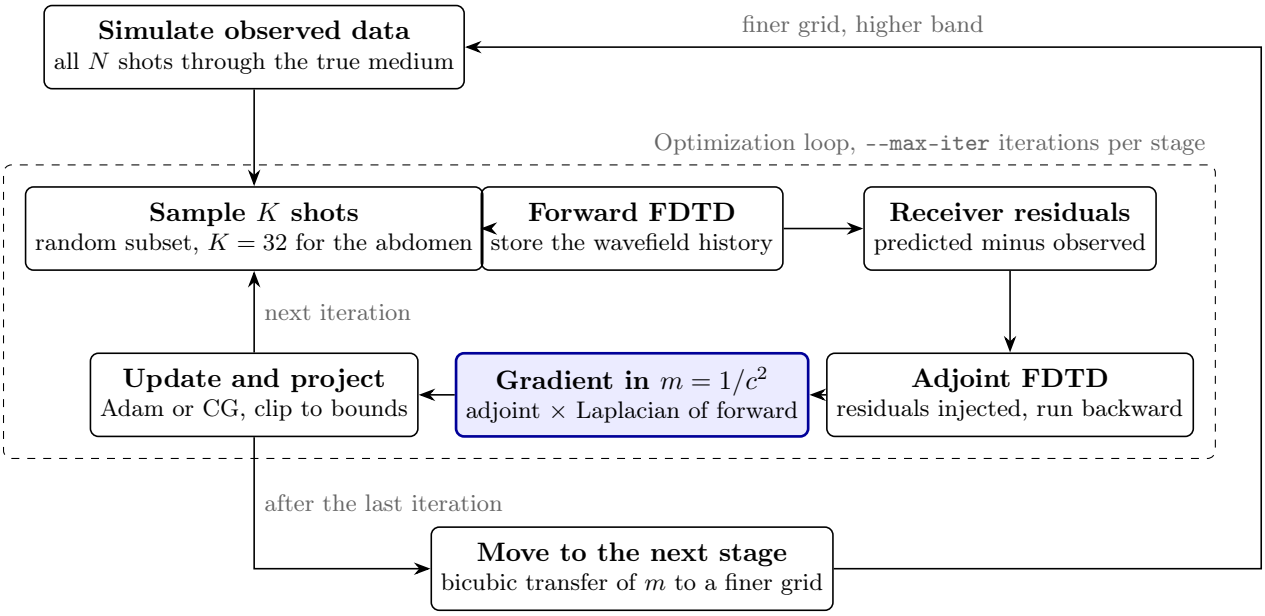


Figure 2: FWI loop inside each multiscale stage. Every iteration costs one forward and one adjoint solve per sampled shot; in the abdomen run the two stages repeat the whole loop at 10–50 and 50–125 kHz.

5 Implementation and verification

The adjoint gradient matches finite differences to second order, and the cost of one iteration is two wave solves per sampled shot plus the stored wavefield.

- **Batching.** `TorchFDTD2DBatch` advances independent shots as one (batch, n_y, n_x) tensor. Each forward step is fused with the receiver gather, and each adjoint step with the residual injection

and the gradient accumulation, so a time step is one compiled call (`torch.compile(mode="reduce-overhead")` on CUDA). Against the earlier unfused code this is about 9×, 6× and 2× faster per shot on the three abdomen grids.

- **Memory.** The forward pass keeps the wavefield of every shot in the batch for the adjoint: $n_{\text{steps}} \times n_y \times n_x \times 4$ bytes per shot in float32 (1.95 GB on a 401^2, 3039-step grid). Storing every fourth step in float16 cuts this 8×; the two-stage abdomen run peaks at about 10 GB.
- **Gradient check.** `python invert_ring.py --gradient-check --dtype float64` compares the adjoint directional derivative $g \cdot \delta m$ with central differences on a 41×41 grid with a 12 mm, 1800 m/s inclusion (Table 5).

Table 5: Adjoint gradient check (float64, run on September 28, 2026).

Step ε	Relative error
10^{-2}	3.5×10^{-3}
10^{-3}	3.4×10^{-5}
10^{-4}	3.4×10^{-7}

The error falls 100× per 10× smaller step, the expected behaviour for a correct gradient with a second-order difference; the approximate adjoint of the Mur boundary does not show at this level. A pytest suite (`tests/`) checks the neural parameterization's chain rule the same way on ring and arc problems, and checks that the fused batched gradient matches an unbatched reference to 10^{-12}, the subsampled and float16 gradients, and the TV gradient against finite differences.

6 Abdomen imaging results

Two-stage FWI of the segmentation-based s1366 abdomen, started from uniform 1480 m/s water, recovers the soft-tissue anatomy: the interior RMS error falls from 297.2 to 142.6 m/s, the soft-tissue RMS error from 53.6 to 36.9 m/s, and to 5.7 m/s more than 10 mm from bone. The remainder is under-recovered bone and the tissue immediately around it. Section 6.2 records the runs that led to this setup, including why the 401^2 stage is dropped.

6.1 Segmentation-based abdomen (TotalSegmentator s1366)

The run used the s1366 map of Section 2.2 and the setup of Table 1, on an NVIDIA RTX A6000 (desmond):

```
python invert_ring.py --device cuda --compile on --no-show \
  --ct-speed totalseg_s1366_acoustic.npz \
  --multiscale --coarse-band 10 50 --n-elements 128 --n-shots 32 \
  --max-iter 200 200 --adam-lr 1e-8 4e-9 --reg 0 --tv 0 1e7 \
  --store-every 4 --wave-dtype float16 \
  --c-min 1400 --c-max 3900 --save-model totalseg_s1366_final200_model.npz \
  --save-figure totalseg_s1366_final200_multiscale.png \
  --display-range 1420 1620 --display-diff 300 --crop-to-ring
```

`--multiscale` runs two stages by default. The coarse stage is unregularized and uses the larger step; stage 2 adds TV with $\beta = 10^7$. Both stages run 200 iterations: with 50, stage 2 had not converged

(Table 8). The log is `totalseg_s1366_final200_log.txt`; the models are `totalseg_s1366_final200_model.npz` (stage 2) and `totalseg_s1366_final200_model_stage1.npz`.

Table 6: Final s1366 run per stage (32 random shots per iteration, one GPU batch). J at iteration 1 is after the first update. Times include the observed-data simulation.

Stage	Iterations	J, iteration 1	J, final	RTX A6000 (s)
101^2, 4.00 mm, 10–50 kHz	200	1.46×10^5	3.91×10^1	76
201^2, 1.83 mm, 50–125 kHz	200	7.97×10^4	1.88×10^3	311
Total				386

Table 7: Interior speed error by tissue for the final s1366 run, computed from the saved stage-2 model on the 201^2 grid. "Start" is the uniform 1480 m/s starting model.

Region	Pixels	Start RMS (m/s)	Final RMS (m/s)	Final bias (m/s)
All interior	25 493	297.2	142.6	−11
All soft tissue ($c < 1700$ m/s)	25 070	53.6	36.9	+2
more than 10 mm from bone	22 780		5.7	
within 10 mm of bone	2 290		120.7	
Fat	5 699	30.0	8.3	+3
Soft-tissue average (other, bowel, pancreas)	5 823	60.0	75.5	+10
Muscle	769	100.0	8.0	−3
Spleen	138	87.0	32.3	−15
Kidney	368	80.0	6.4	−3
Liver	3 915	90.0	7.4	−1
Blood vessels	349	95.0	7.1	−5
Water, bile, urine (incl. coupling water)	8 009	0.0	2.5	0
Bone	423	2270.0	1069.9	−783

- **Soft tissue.** Liver, kidney, spleen, muscle, vessels and the subcutaneous and visceral fat appear with sharp boundaries and nearly unbiased speeds; away from bone the error map is blank except for faint lines along tissue boundaries (Figure 3).
- **Near bone.** The soft tissue within 10 mm of bone carries almost all the remaining soft-tissue error (120.7 m/s RMS over 9% of the soft-tissue pixels), most visibly the tissue enclosed by the vertebral body, which comes out too fast. This is why the "soft-tissue average" class, which includes the paravertebral tissue, ends worse than it started, and why the soft-tissue total rises slightly between 50 and 200 iterations (32.2 to 36.9 m/s) while every organ improves.
- **Bone.** The vertebral body, spinous process and every rib are in the right place, but too slow: median 2730 against 3750 m/s, up from 2210 after 50 iterations. Some bone pixels reach the 3900 m/s bound.
- **Artifacts.** No speckle or radial streaks: at 125 kHz the 8.3 mm element pitch is 0.7 wavelengths, close enough to adequate sampling.
- **Convergence.** The coarse stage fits its data almost exactly (J falls 3700×). Stage 2 falls 42× and has nearly levelled off: J drops 62% between iterations 50 and 100, 16% from 100 to 150 and 3% from 150 to 200.

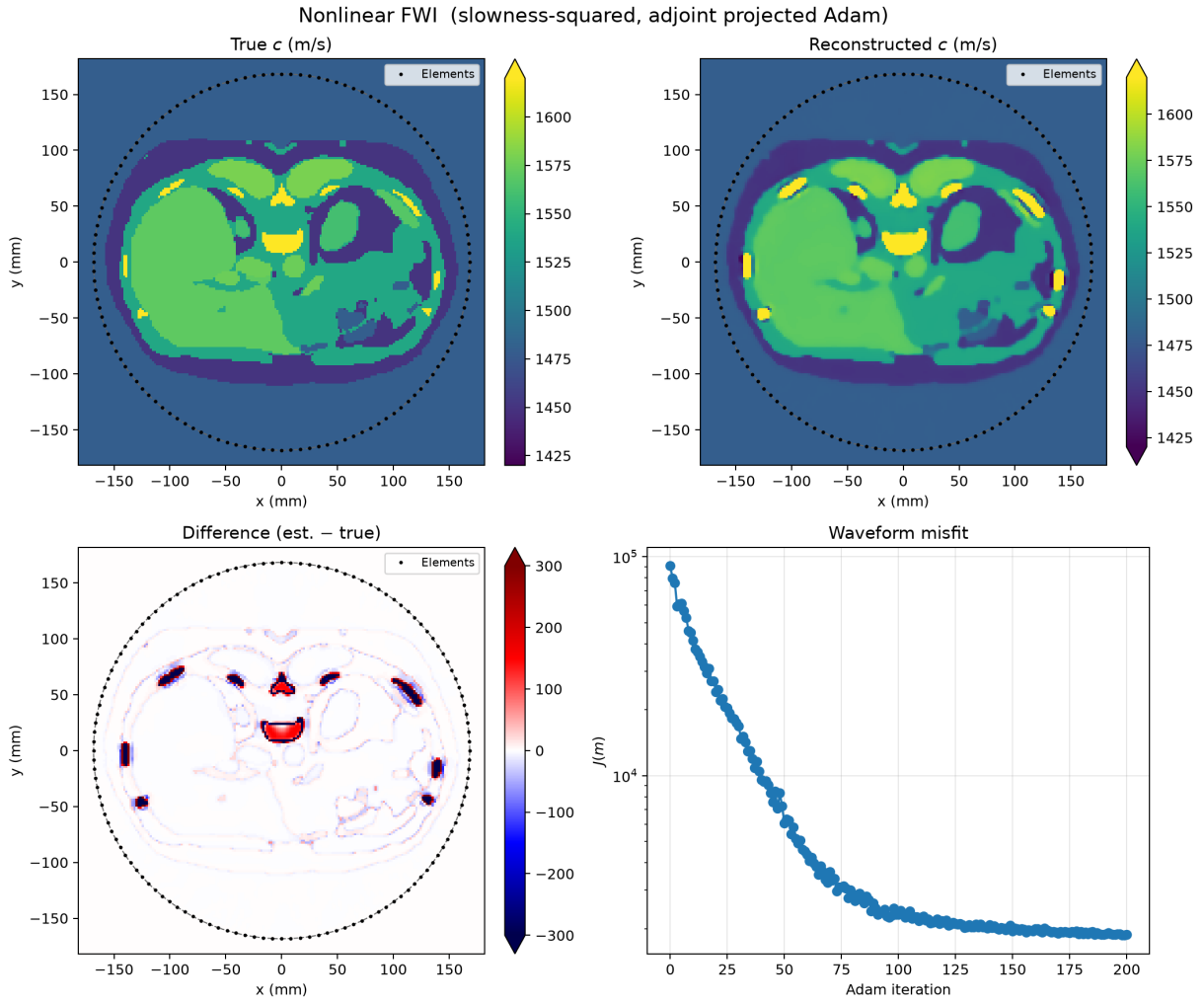


Figure 3: Two-stage FWI of the s1366 abdomen, cropped to the ring. True c and the stage-2 reconstruction in a 1420–1620 m/s window (bone saturates), difference clipped at ±300 m/s, and the stage-2 misfit. The plot's y axis runs anterior (bottom) to posterior (top).

6.2 How the setup was reached

Table 8 lists the s1366 runs in order. All used 128 elements and started from uniform water unless noted. Until the last three rows, every run included the 401^2, 100–250 kHz stage, and the errors are those of its output.

- **A batching bug.** Batched runs never excluded the transmitter's own channel from the misfit: the zeroing acted on a copy made by advanced indexing. Every run before the third row is affected, including the KiTS run (Section 6.3). It was found by a test comparing the batched gradient with an unbatched reference.
- **Shots and elements.** More shots per iteration did not remove the artifacts. 256 elements removed the radial streaks near the ring, confirming that 128 elements are too sparse at 250 kHz, but the interior did not improve.

Table 8: s1366 runs on the RTX A6000. RMS errors in m/s, interior / soft tissue ($c < 1700$ m/s). Uniform water scores 297 / 54 on the 201^2 grid.

Run	Final model	Wall time
4 shots, no regularization, 25–75 kHz coarse band (original code)	315 / 204	6 min
64 shots (original code)	338 / 252	44 min
32 shots, fused code, transmitter-channel fix, Tikhonov 10^{14}	314 / 216	7 min
+ 10–50 kHz coarse band, Tikhonov 10^{15}	288 / 170	7 min
+ 256 elements on stages 2–3	304 / 195	7 min
128 elements, coarse stage 200 iterations at 10^{-8}	293 / 180	8 min
Tikhonov 0 / 10^{14} / 10^{14}	353 / 268	8 min
Same, started from the true map blurred by 10 mm	350 / 265	8 min
TV 0 / 10^7 / 10^7, stage-3 output	353 / 269	8 min
Same run, stage-2 model	**185 / 32**	—
Two stages, 50 stage-2 iterations	185 / 32	2.7 min
Two stages, 200 stage-2 iterations (Section 6.1)	**143 / 37**	6.4 min

- **Regularization.** Tikhonov at 10^{15} suppressed speckle but charged the true model's bone edges about 10^4 on the coarse grid, six times the converged coarse misfit, so it blocked the bone. TV charges jumps linearly and is used instead on stage 2.
- **The decisive diagnostic.** Saving the model after every stage showed that stages 1 and 2 work and stage 3 fails (Figures 4 and 5). From uniform water and from a 10 mm-blurred true map alike, the stage-2 model has 185–187 / 32–33 m/s error; stage 3 turns both into the same speckled, biased model at about 350 / 265 m/s. The output of stage 3 does not depend on its input.

6.3 Earlier experiment: KiTS-00136 with the HU-threshold model

The first abdomen runs used the `ct_to_speed.py` map of C4KC-KiTS case KiTS-00136 (categorical mapping, 1478–3476 m/s). Its stages ran at 4.63, 2.12 and 1.01 mm on 468, 425 and 406 mm fields of view, with a 194.4 mm ring of 64 elements (19 mm pitch). Three-stage FWI recovered the soft-tissue anatomy and the outer cortex of the vertebra, but left the vertebral body hollow, underestimated by up to about 2000 m/s.

6.3.1 Full multiscale reconstruction

- **Soft tissue.** Organ boundaries, the paraspinal muscles, the abdominal-wall muscle and the fat layers appear in the right places with roughly the right speeds; the difference map is near zero across most of the abdomen.
- **Bone.** The cortical rim of the vertebral body and the posterior elements are recovered, partly at the full 3476 m/s. The interior of the vertebral body stays near the background speed, the dark region in the difference map.
- **Artifacts.** Faint radial streaks fan inward from the ring. With 64 elements the pitch is about 19 mm, several wavelengths at 100–250 kHz, so the source and receiver sampling is sparse at the finest stage.

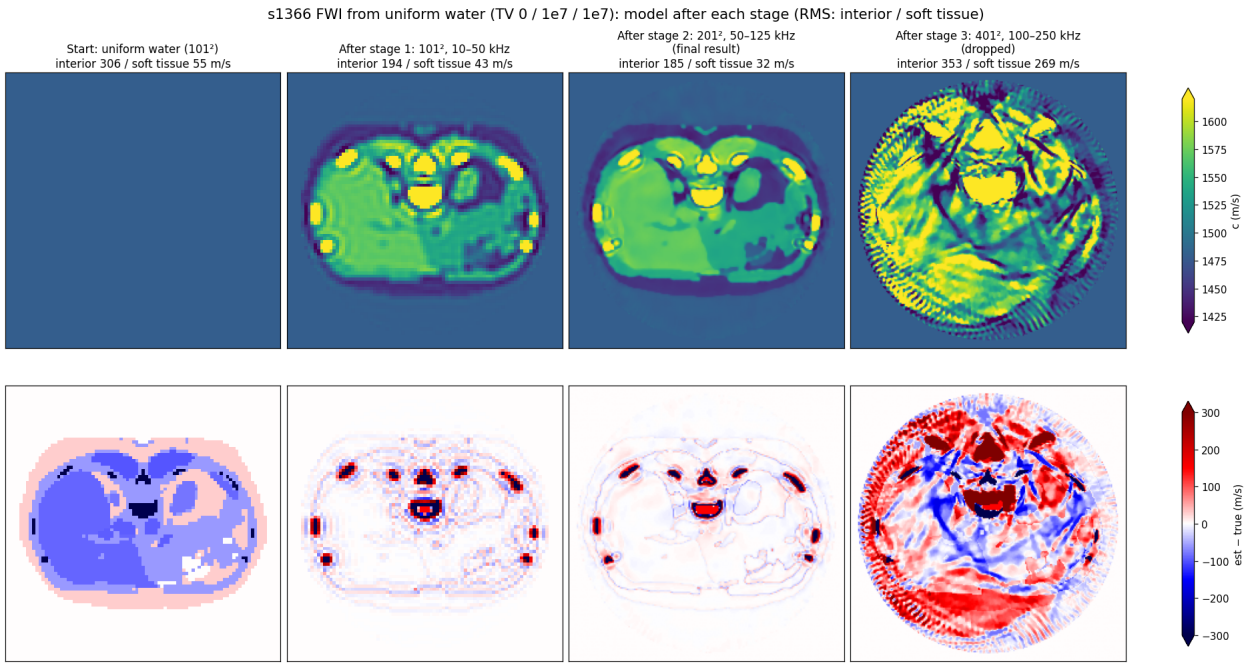


Figure 4: Run with TV 0 / 10^7 / 10^7 started from uniform water: speed (top) and error (bottom) at the start and after each stage. Stage 2 is the result; stage 3 destroys it.

- **Convergence.** In the final stage J falls from 5.81×10^3 to 5.78×10^2 (10×), with a rise after the first step and the noise expected from random shot sub-sampling.

The run was made on the RTX A6000 (desmond) on September 26, 2026, with

```
python invert_ring.py --device cuda --ct-speed abdomen_acoustic.npz \
  --multiscale --c-max 3600 --n-elements 64 --n-shots 16 \
  --parallel-shots 16 --adam-lr 5e-09 --max-iter 50 \
  --no-show --save-figure abdomenmultiscale.png
```

(recovered from the shell history on desmond; the original line also carried a superseded `--save-figure abdomen_inverse.png` and a repeated `--no-show`). The output is in `abdomenmultiscale_log.txt`, and the figure on desmond is byte-identical to the committed one. All other options took the defaults of that version of the code: no regularization, $c_{\min} = 1400$ m/s, shot seed 0, float32 with a compiled step. Like every batched run before commit `d519e92`, it kept each transmitter's own channel in the misfit. Table 9 lists the logged numbers.

6.3.2 Coarse-stage reproduction

Before the multiscale log was recovered, the first stage was rerun on CPU on October 27, 2026 to inspect the coarse result on its own. This rerun used 4 gradient shots and a learning rate of 4×10^{-9} rather than the original 16 shots and 5×10^{-9}:

```
python invert_ring.py --ct-speed abdomen_acoustic.npz --coarse-only \
  --n-elements 64 --n-shots 4 --max-iter 50 --adam-lr 4e-9 \
  --c-min 1400 --c-max 3600
```

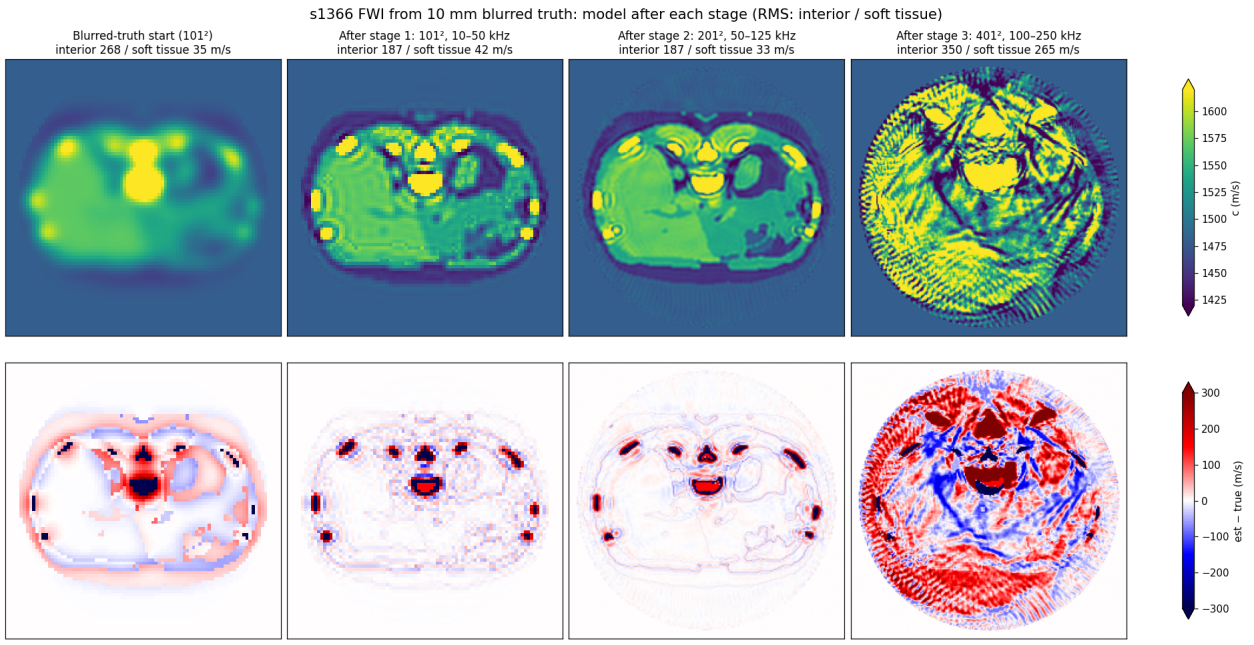


Figure 5: The same diagnostic from the true map blurred by a 10 mm Gaussian (Tikhonov 0 / 10^{14} / 10^{14}). Stages 1 and 2 converge to the same model as the water start; stage 3 again degrades it.

Table 9: KiTS-00136 multiscale run, from `abdomenmultiscale_log.txt`. J at iteration 1 is after the first Adam update.

Stage	Grid, spacing, band	J at iteration $1 \to 50$	Wall time
1	101^2, 4.63 mm, 25–75 kHz	$2.34 \times 10^4 \to 6.33 \times 10^3$	88 s
2	201^2, 2.12 mm, 50–125 kHz	$9.28 \times 10^3 \to 1.59 \times 10^3$	191 s
3	401^2, 1.01 mm, 100–250 kHz	$7.82 \times 10^3 \to 5.78 \times 10^2$	432 s

Elements 64; gradient shots 16 of 64 per iteration; Adam learning rate 5×10^{-9}
Final stage: 4704 steps at $\Delta t = 89.6$ ns; total wall time 711 s
Interior RMS error in c: 223.7 m/s; reconstructed range 1400–3600 m/s (true 1478–3476)

The full log is in `abdomen_coarse_stage_log.txt`.

The soft-tissue structures appear after a single coarse stage, but the vertebra goes the wrong way: speed falls to the lower bound where it should rise to 3476 m/s. A straight path through about 40 mm of bone arrives about 15 μs earlier than through water, more than half the 13 μs period at 75 kHz, so the coarse data are already cycle-skipped there. The later stages start from this hollow vertebra and only sharpen its rim, which matches Figure 6. The RMS error is dominated by the bone pixels; the difference map is close to zero over most of the soft tissue.

7 Limitations and future work

- **The 100–250 kHz stage (future research).** Run from the stage-2 model, the 401^2 stage cycle-skips: its starting misfit is no lower than from far worse models, and it converges to the same wrong model whatever its input. At 250 kHz a half period is 2 μs; the stage-2 model's residual errors (about 30 m/s along 200 mm paths, plus the bone) already exceed it. Directions:

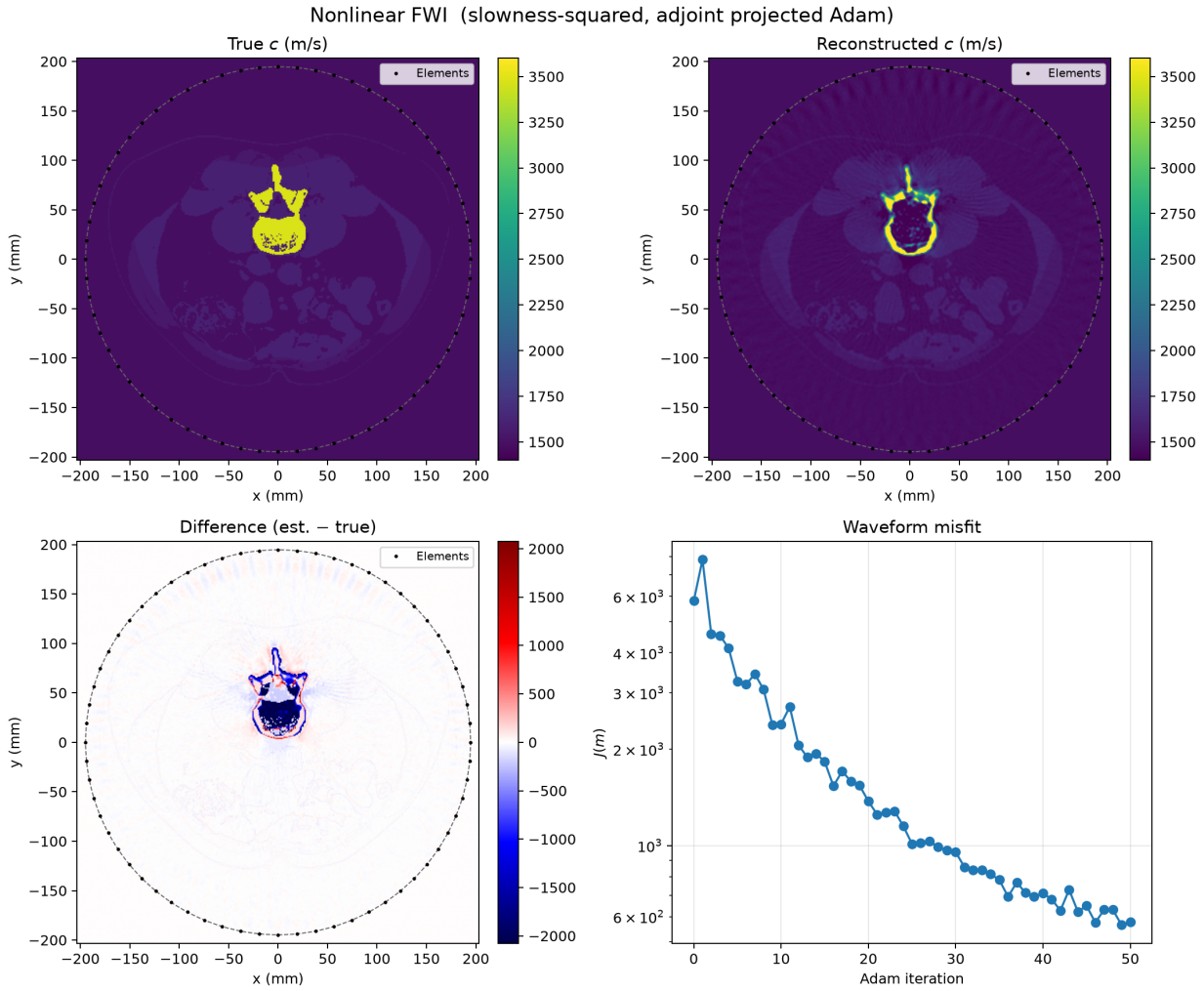


Figure 6: Multiscale FWI of the abdomen (`abdomenmultiscale.png`, committed September 26, 2026). True c, reconstruction after the 401^2 / 100–250 kHz stage, difference, and the final-stage waveform misfit over 50 Adam iterations.

smaller frequency steps toward 250 kHz (for example 100–150, 125–200 and 150–250 kHz); a smaller step size, or L-BFGS with illumination-compensated gradients, since Adam moves every pixel by about its step size even where the gradient is noise; more elements (256 removed the near-ring streaks); and a fourth-order stencil, since the 401^2 s1366 grid has only 3.9 points per shortest wavelength.

- **Bone.** Bone is located but reaches only about 70% of its speed, and the soft tissue within 10 mm of it is off by about 120 m/s. Options: a travel-time or envelope misfit, a bone-aware prior (the segmentation marks the bone), or inverting soft tissue with the bone held fixed.

- **Inverse crime.** Observed and predicted data come from the same solver on the same grid, with no noise. Results are optimistic; data should be generated on a finer grid, or with an independent solver, plus noise.

- **Starting model and sampling.** The blurred-truth test shows the two-stage result does not

Table 10: Coarse-stage abdomen reproduction.

Quantity	Value
Grid / band	101×101 at 4.63 mm, 25–75 kHz
Time steps, Δt	1029, 409 ns
Misfit J, start $\rightarrow$ end	$7.35 \times 10^3 \rightarrow 1.35 \times 10^3$ (5.4× lower)
Interior RMS error in c	298.5 m/s
Interior c range, true	1478–3476 m/s
Interior c range, reconstructed	1400–1832 m/s
Wall time (8 CPU threads, float32)	71 s (observed data 2.1 s, optimization 68.6 s)

depend on the starting model for this slice. The 128-element ring is adequately sampled up to about 125 kHz; higher frequencies need more elements.

- **Physics.** Constant density, no attenuation and no shear waves. Bone and gas are only approximated by a scalar acoustic model. The density and attenuation maps written by `totalseg_to_speed.py` are not yet used by the solvers.
- **Reproducibility.** The commands and logs of all abdomen runs are saved alongside their figures. The KiTS-00136 command was recovered after the fact from shell history.

Sources

- Repository scjoshi/SoundWaveSimulation: `invert_ring.py`, `fdtd2d/torch_backend.py`, `fdtd2d/inversion/` (CG, misfit, medium), `fdtd2d/ct_medium.py`, `ct_to_speed.py`, `fdtd2d/inversion/neural.py`.
- Final s1366 run: `totalseg_s1366_final200_multiscale.png`, `totalseg_s1366_final200_log.txt` and the saved models; earlier runs in Table 8 as `totalseg_s1366_*` with their logs; converter `totalseg_to_speed.py` and `abdominal_acoustic_parameters.csv`.
- Abdomen figure `abdomenmultiscale.png`, commit `2de3791` (September 26, 2026), and its log `abdomenmultiscale_log.txt`; CT map figure `abdomen_acoustic.png`.
- CT slice: `ct-abdomen-c4kc-kits-instance.dcm`, C4KC-KiTS collection, The Cancer Imaging Archive.
- Coarse-stage reproduction: run on October 27, 2026; gradient check: run on September 28, 2026 (both CPU, PyTorch 2.6).

References

[1] F. A. Duck, *Physical Properties of Tissue: A Comprehensive Reference Book*. London: Academic Press, 1990.

[2] T. L. Szabo, *Diagnostic Ultrasound Imaging: Inside Out*, 2nd ed. Elsevier Academic Press, 2014.

[3] T. D. Mast, "Empirical relationships between acoustic parameters in human soft tissues," *Acoustics Research Letters Online*, vol. 1, no. 2, pp. 37–42, 2000.

[4] J. Wasserthal *et al.*, "TotalSegmentator: robust segmentation of 104 anatomic structures in CT images," *Radiology: Artificial Intelligence*, vol. 5, no. 5, e230024, 2023.

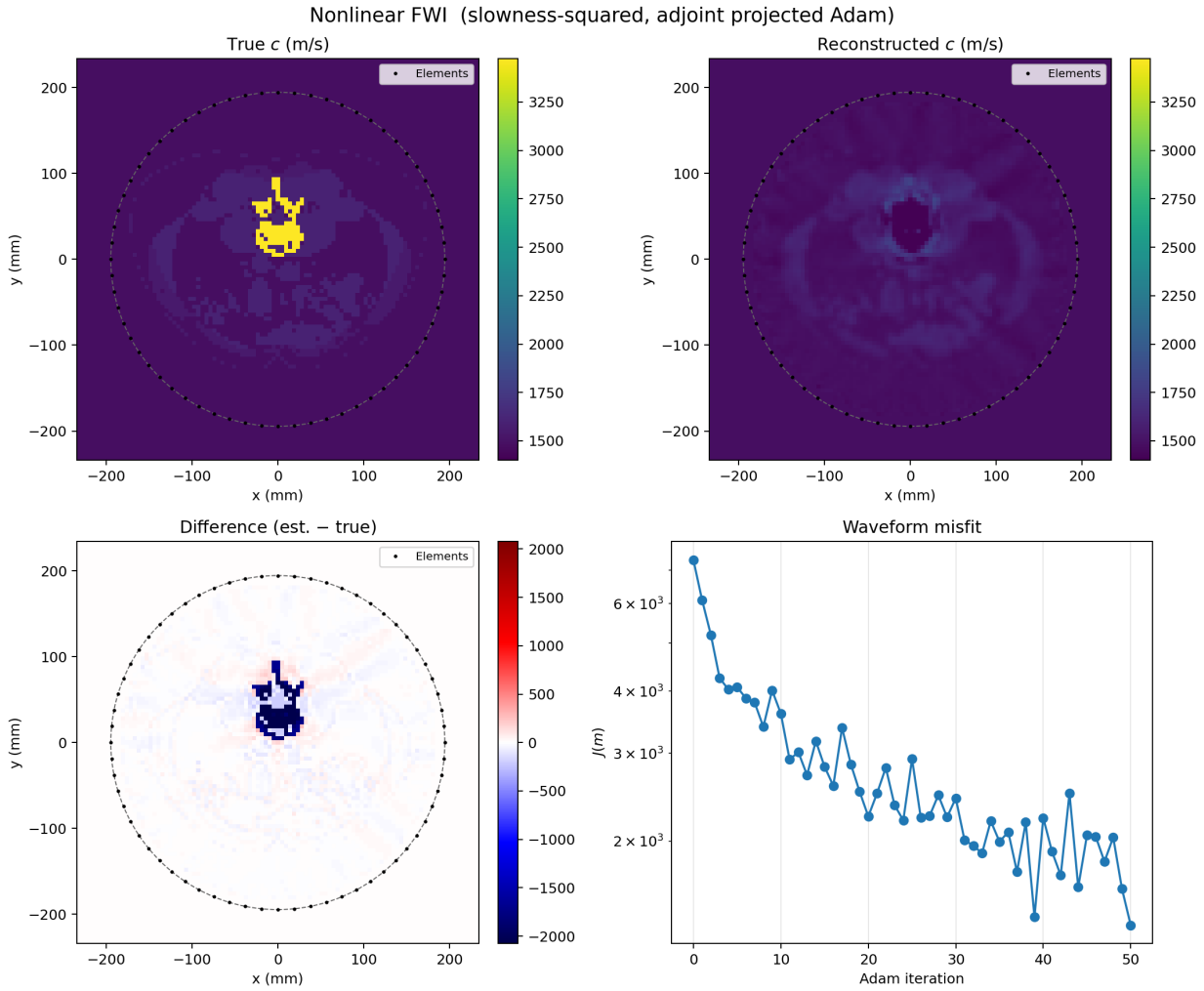


Figure 7: Coarse stage only (101^2, 25–75 kHz, 50 Adam iterations). Organ and muscle outlines are already visible; the vertebra region is driven down to the 1400 m/s lower bound.

[5] J. Wasserthal, "Small subset of TotalSegmentator Dataset," version 2.0.1, Zenodo, 2023, https://zenodo.org/records/10047263 (CC BY 4.0).

[6] M. O. Culjat, D. Goldenberg, P. Tewari, and R. S. Singh, "A review of tissue substitutes for ultrasound imaging," *Ultrasound in Medicine & Biology*, vol. 36, no. 6, pp. 861–873, 2010.